

Liangliang Zhang

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EDUCATION

Rensselaer Polytechnic Institute, Troy, USA

August 2024 - May 2028

Ph.D. in Computer Science advised by Prof. Yao Ma

Central South University, Changsha, China

September 2022 - June 2024

B.S. in Software Engineering

RESEARCH INTEREST

Large Language Models, GraphRAG, Knowledge Graphs, Graph Neural Networks, Trustworthy AI, Data-Centric AI, Data Condensation.

SELECTED HONORS AND AWARDS

- **Best Doctoral Consortium Presentation Award, WSDM 2026** Feb 2026
Awarded for the top presentation at the Doctoral Consortium for the paper: “Planning Before Retrieval: A Constrained Planner-Executor Framework for Reliable and Efficient KG-Augmented Reasoning.”
- **WSDM 2026 Student Travel Award** Feb 2026
Competitive travel grant awarded for contributions and participation at WSDM 2026.
- **KDD 2024 Student Travel Award** Jun 2024
Competitive travel grant awarded for contributions and participation at KDD 2024.
- **Academic Year Scholarship, School of Computer Science, Central South University**
Awarded four times for academic excellence and outstanding performance.

CONFERENCE AND JOURNAL PUBLICATIONS

- [1] **Zhang, Liangliang** and Y. Ma. “Planning Before Retrieval: A Constrained Planner-Executor Framework for Reliable and Efficient KG-Augmented Reasoning”. In: *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining*. 2026, pp. 1308–1310.
- [2] **Zhang, Liangliang**, Z. Jiang, H. Chi, H. Chen, M. Elkoumy, F. Wang, Q. Wu, Z. Zhou, S. Pan, S. Wang, et al. “Diagnosing and Addressing Pitfalls in KG-RAG Datasets: Toward More Reliable Benchmarking”. In: *In Proceedings of the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS)* (2025).
- [3] **Zhang, Liangliang**, H. Bao, and Y. Ma. “Extending Graph Condensation to Multi-Label Datasets: A Benchmark Study”. In: *Transactions on Machine Learning Research (TMLR)* (2025).
- [4] H. Xu, **Zhang, Liangliang**, Y. Ma, S. Zhou, Z. Zheng, and J. Bu. “Learning to Reduce the Scale of Large Graphs: A Comprehensive Survey”. In: *ACM Transactions on Knowledge Discovery from Data* 19.5 (2025), pp. 1–25.
- [5] Zhao, Tianyi*, **Zhang, Liangliang***, Y. Ma, and L. Cheng. “A Survey on Safe Multi-Modal Learning System”. In: *In Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)* (2024). * contributed equally to this work.

- [6] **Zhang, Liangliang**, M. Chen, X. Hu, and L. Deng. “Graph Convolutional Network and Contrastive Learning Small Nucleolar RNA (snoRNA) Disease Associations (GCLSDA): Predicting snoRNA–Disease Associations via Graph Convolutional Network and Contrastive Learning”. In: *International Journal of Molecular Sciences* 24.19 (2023), p. 14429.
- [7] D. Liu, X. Li, **Zhang, Liangliang**, X. Hu, J. Zhang, Z. Liu, and L. Deng. “Hgnnlda: predicting lncrna-drug sensitivity associations via a dual channel hypergraph neural network”. In: *IEEE/ACM transactions on computational biology and bioinformatics* 20.6 (2023), pp. 3547–3555.

PREPRINTS AND SUBMISSIONS

- [8] **Zhang, Liangliang**, N. Mihindukulasooriya, N. S. D’Souza, S. Shirai, S. Dash, Y. Ma, and H. Samulowitz. “From Factoid Questions to Data Product Requests: Benchmarking Data Product Discovery over Tables and Text”. In: *arXiv preprint arXiv:2510.21737* (2025).
- [9] H. Xu, **Zhang, Liangliang**, Y. Ma, S. Zhou, Z. Zheng, and B. Jiajun. “A Survey on Graph Condensation”. In: *arXiv preprint arXiv:2402.02000* (2024).

TEACHING & RESEARCH EXPERIENCE

DAMI Lab | *Research Assistant* January 2025 – Present
 Research on Data-Centric AI (Data Augmentation, Active Learning, Data Valuation, Data Condensation) and Large Language Models (GraphRAG, Knowledge Graphs for LLMs).

RPI CSCI 1100: Computer Science, | *Teaching Assistant* August 2024 – December 2024
 Graded exams and assignments, proctored tests, led lab sessions, and held office hours to support student learning (20 hrs/week).

Competitive Programming Club | *Team Member* July 2022 – September 2023
 Research on Data Mining and Graph Neural Networks and Bioinformatics.

Big Data Computing | *Team Member* September 2021 - September 2022
 Researched and implemented swarm intelligence algorithms for resource scheduling.

INTERNSHIP

IBM Yorktown Heights May - August 2025

- Developed a large-scale data product retrieval benchmark integrating structured tables and unstructured text for enterprise data discovery.
- Built a multi-vector hybrid retrieval framework unifying semantic and lexical representations via rank fusion, achieving high recall and robustness at scale.
- Established reproducible evaluation protocols for metadata-only retrieval, supporting scalable and consistent benchmarking.

Institute of Automation, Chinese Academy of Sciences March - September 2023

- Enhanced skills in Linux and Docker for system operations.
- Debugged deep learning interfaces and developed attack algorithms.
- Optimized CW attacks and implemented AutoAttack and CopyCAT methods.

Hunan Wuche Education Technology Co., Ltd. August - September 2022

- Co-designed an online tank battle game using JavaFX and FXGL.
- Led project design and presentations, enhancing Java expertise in UI, SQL, and Javax.mail.

China Hi-Tech Group

June - July 2022

- Developed a shopping mall app with Vue and SpringBoot.
- Directed program design, task coordination, and team meetings.
- Gained proficiency in SpringBoot, Vue, and JavaScript.

Guangdong QST Green Soft Training Education Technology Co., Ltd.

June - July 2021

- Created an online Gomoku game in C++.
- Implemented user authentication, game modes, and enhanced C++ skills.